

26) Suppose the coefficient matrix of a linear system of three equations in three variables has a pivot in each column. Explain why the system has a unique solution.

Soln: The coefficient matrix has a size $[3 \times 3]$. Since each column has a pivot, there are at least 3 pivots. By defn. of a pivot position, it corresponds to a leading 1 in the reduced row echelon form of a matrix. By defn. of a leading 1, it's logical that a single row must have exactly one pivot. $\therefore$ Each of the 3 rows has a single pivot each. $\therefore$ No row is non-zero. The augmented matrix has a size 3×4 . Since each row has a single pivot and there are only 3 pivots, column 4 is not a pivot column. Since no row of the matrix in augmented form has the form $[0 \dots 0 \ b]$, with $b \neq 0$, the system is consistent.

Each of the 3 columns of the coefficient matrix are pivot columns. $\therefore$ By definition, all three variables are basic variables.

By Theorem 2, the system has a unique solution. $\square$

27) Restate the last sentence in Theorem 2 using the concept of pivot columns: "If a linear system is consistent, then the solution is unique if and only if every column in the coefficient matrix is a pivot column."

28) What would you have to know about the pivot columns in an augmented matrix in order to know that the linear system is consistent and has a unique solution?

Soln: From the solution of Q 24, we know that in order for a system to be consistent, the last column of the augmented matrix should not be a pivot column. From the solution of Q 27, we know that a linear system is consistent and has a unique solution if and only if every column in the coefficient matrix is a pivot column. $\therefore$ In the corresponding augmented matrix, every column apart from the last column must be a pivot column.